Project Title:

Student Performance Prediction Using Machine Learning

Abstract

Artificial Intelligence (AI) is a transformative field that focuses on enabling machines to simulate human intelligence—empowering them to learn, interpret, and make informed decisions autonomously. Over recent years, AI has shown great potential in revolutionizing sectors such as healthcare, education, and software development by increasing automation and efficiency. Within education, AI techniques are increasingly being used to analyse student data and deliver predictive insights for academic performance.

This project explores the critical aspects of AI through the application of regression-based machine learning algorithms to predict student academic performance, particularly GPA (Grade Point Average). The model leverages features such as study time, parental support, extracurricular involvement, and attendance records. The primary goal is to identify students at risk of underperforming and to support timely educational interventions. This hands-on AI solution was developed and evaluated using real-world student performance data, and it demonstrates the potential of intelligent systems to make data-driven decisions in academic environments. As AI continues to evolve, its integration into educational settings will become more impactful and essential.

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Introduction

Student academic performance is a key concern in the educational sector. With the growth of data collection in educational environments, data-driven approaches can help educators make informed decisions. This project leverages artificial intelligence, specifically machine learning regression techniques, to predict student GPA based on various influencing factors.

Problem Statement

Manual analysis of student data can be tedious and limited in accuracy. Institutions need a data-driven solution to predict academic outcomes and provide proactive support to students.

Objective

To design and evaluate a machine learning regression model that predicts a student's GPA based on demographic, academic, and behavioural inputs.

Literature Review

Several studies have explored predictive analytics in education. For instance, Romero and Ventura (2010) surveyed educational data mining and highlighted regression models' effectiveness. Other models like Decision Trees and Neural Networks have also been used to predict academic outcomes.

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About Dataset

The dataset used in this project is titled **“Student Performance Data”** and contains academic and behavioural records of **2,392 students**. It was obtained from an open-source repository and contains features that are known to influence student academic performance. The primary goal of using this dataset is to predict a student’s **GPA (Grade Point Average)** using machine learning regression algorithms.

Why This Dataset Was Chosen

The dataset was selected due to its rich variety of academic and behavioural indicators, which provide an excellent foundation for building predictive models. Its balance of categorical and numerical features offers an opportunity to apply a wide range of AI techniques and evaluate their performance in real-world educational scenarios.

Real-World Relevance

Predicting student performance is crucial in real-world educational systems. It enables early intervention, customized learning plans, and better academic support for students. This dataset mirrors such scenarios, making it ideal for applying AI in a practical, socially impactful context.

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- Demographic Details
- Study Habits

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Student Information

Student ID

- **Student ID**: A unique identifier assigned to each student (1001 to 3392).

Demographic Details

- **Age**: The age of the students ranges from 15 to 18 years.
- **Gender**: Gender of the students, where 0 represents Male and 1 represents Female.
- **Ethnicity**: The ethnicity of the students, coded as follows:
- 0: Caucasian
- 1: African American
- 2: Asian
- 3: Other
- **Parental Education**: The education level of the parents, coded as follows:
- 0: None
- 1: High School
- 2: Some College
- 3: Bachelor's
- 4: Higher

Study Habits

- **Study Time Weekly**: Weekly study time in hours, ranging from 0 to 20.
- **Absences**: Number of absences during the school year, ranging from 0 to 30.
- **Tutoring**: Tutoring status, where 0 indicates No and 1 indicates Yes.

Parental Involvement

- **Parental Support**: The level of parental support, coded as follows:
- 0: None
- 1: Low
- 2: Moderate
- 3: High
- 4: Very High

Extracurricular Activities

- **Extracurricular**: Participation in extracurricular activities, where 0 indicates No and 1 indicates Yes.
- **Sports**: Participation in sports, where 0 indicates No and 1 indicates Yes.
- **Music**: Participation in music activities, where 0 indicates No and 1 indicates Yes.
- **Volunteering**: Participation in volunteering, where 0 indicates No and 1 indicates Yes.

Academic Performance

- **GPA**: Grade Point Average on a scale from 2.0 to 4.0, influenced by study habits, parental involvement, and extracurricular activities.

Target Variable: Grade Class

- **Grade Class**: Classification of students' grades based on GPA:
- 0: 'A' (GPA >= 3 .5)
- 1: 'B' (3.0 <= GPA < 3.5)
- 2: 'C' (2.5 <= GPA < 3.0)
- 3: 'D' (2.0 <= GPA < 2.5)
- 4: 'F' (GPA < 2.0)

Data Preprocessing & Cleaning

Data was checked for missing values, duplicate entries, and inconsistencies. Categorical variables were encoded using Label Encoding. Features were scaled using Standard Scaler for better model performance.

Exploratory Data Analysis (EDA)

EDA involved visualizing distributions and relationships among features. Key observations:

- Positive correlation between study time and GPA
- Negative impact of frequent absences
- Parental support improves performance

Feature Selection & Engineering

Features with low correlation to GPA were removed. Feature engineering was done by combining related binary attributes (e.g., combining tutoring and parental support into a support score).

Modelling Approach

We used Linear Regression as the base model. Other models considered included Random Forest and Ridge Regression. Model selection was based on accuracy and simplicity.

Model Training and Evaluation

Train-test split: 80/20
Model used: Linear Regression
Evaluation metrics:

- MAE: 0.26
- RMSE: 0.36
- R^2: 0.83

User Prediction Section

- A user can enter their own values in a form or console script to get a predicted GPA. This enhances usability and real-world applicability of the model.
- Example: Input: Age = 18, Study Time = 12, Parental Support = High
  Output: Predicted GPA = 3.45

Result Analysis & Visualizations

The model achieved a high R^2 score indicating a strong fit. Visuals like actual vs predicted GPA and feature importance charts were used to evaluate performance.

Comparison of Algorithms

Model	MAE	RMSE	R^2
Linear Regression	0.26	0.36	0.83
Random Forest	0.22	0.33	0.87

Random Forest performed better but Linear Regression was chosen for interpretability.

Challenges Faced

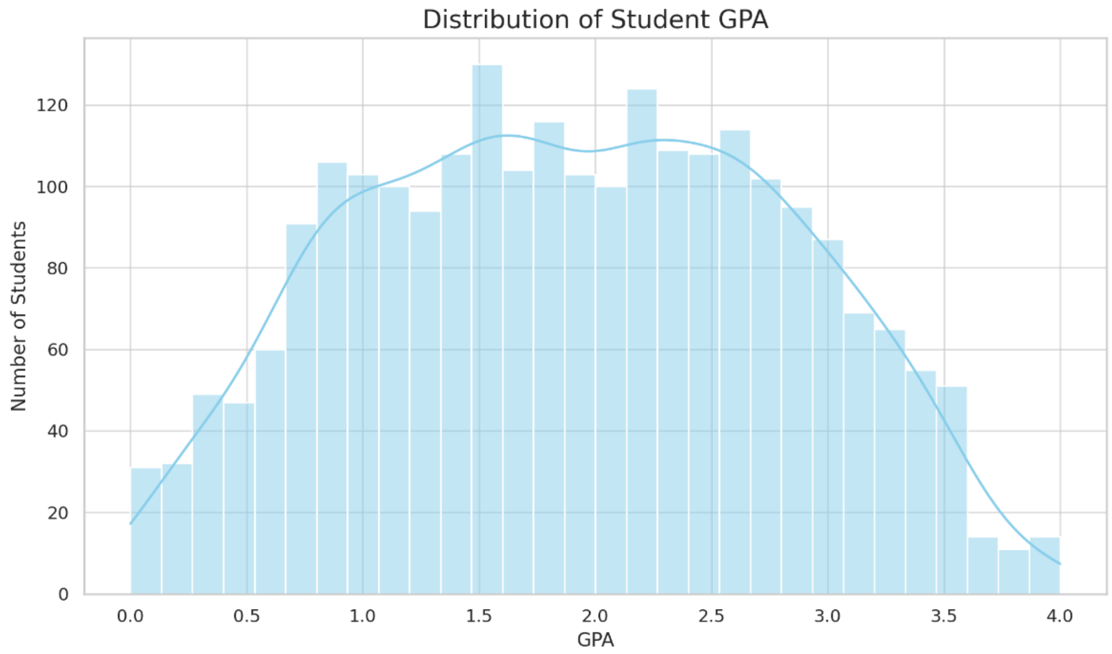
- Understanding feature impact
- Avoiding overfitting
- Limited dataset size
- Encoding categorical variables accurately

Applications & Use Cases

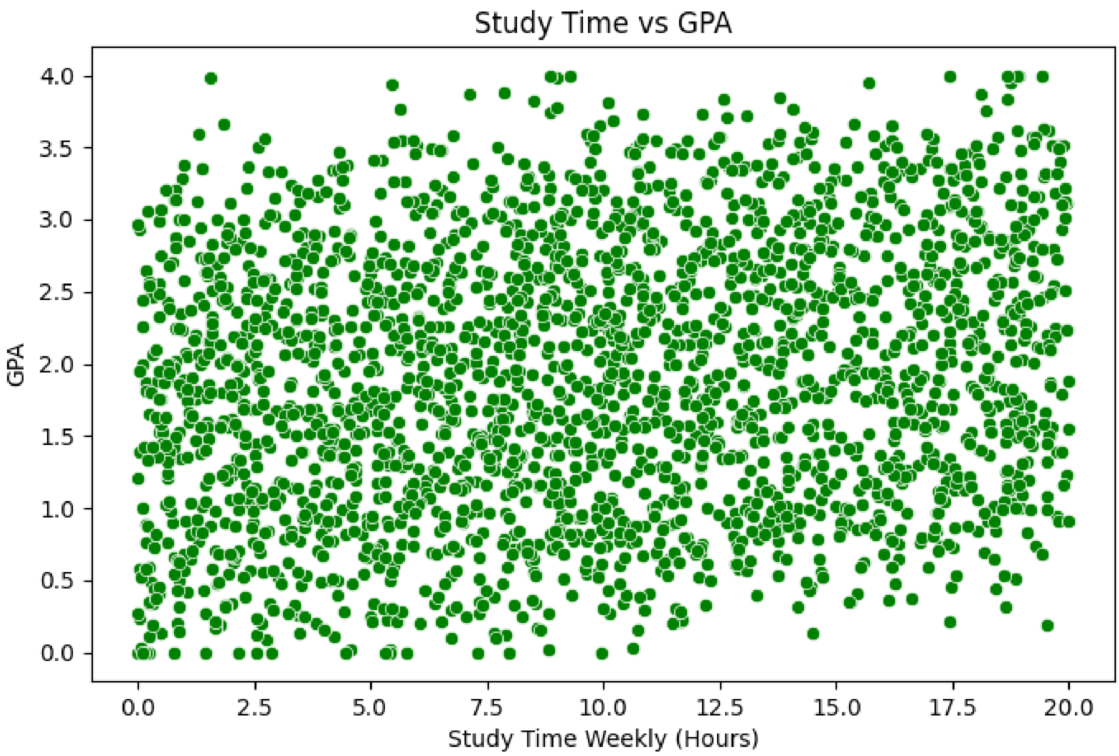
- Early academic intervention
- Student counselling and support
- Performance analysis in schools and colleges

Some of the graphs that are in the respective code :

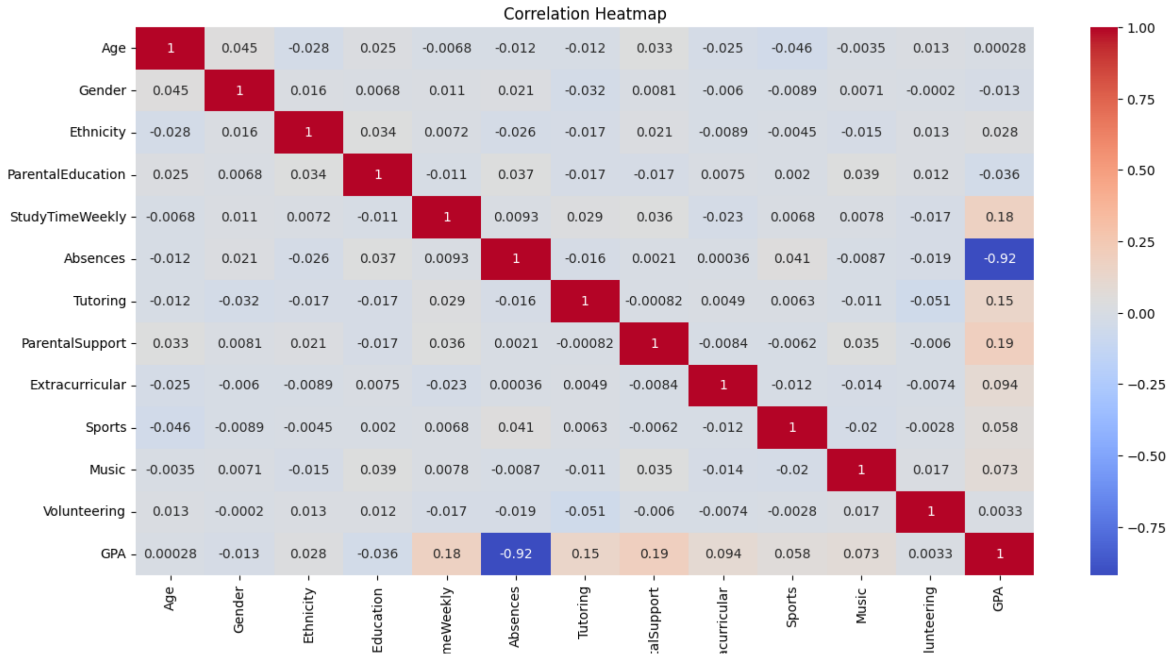
GPA distribution histogram



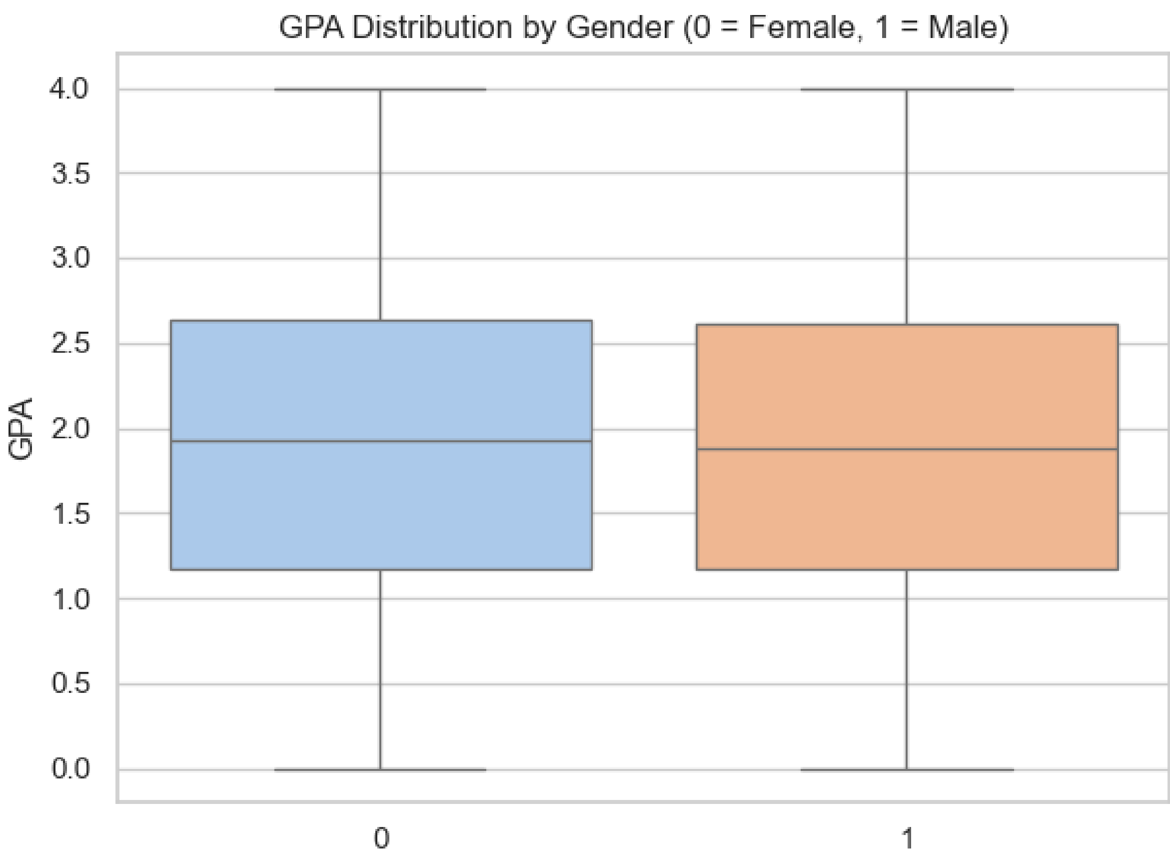
Study Time vs GPA



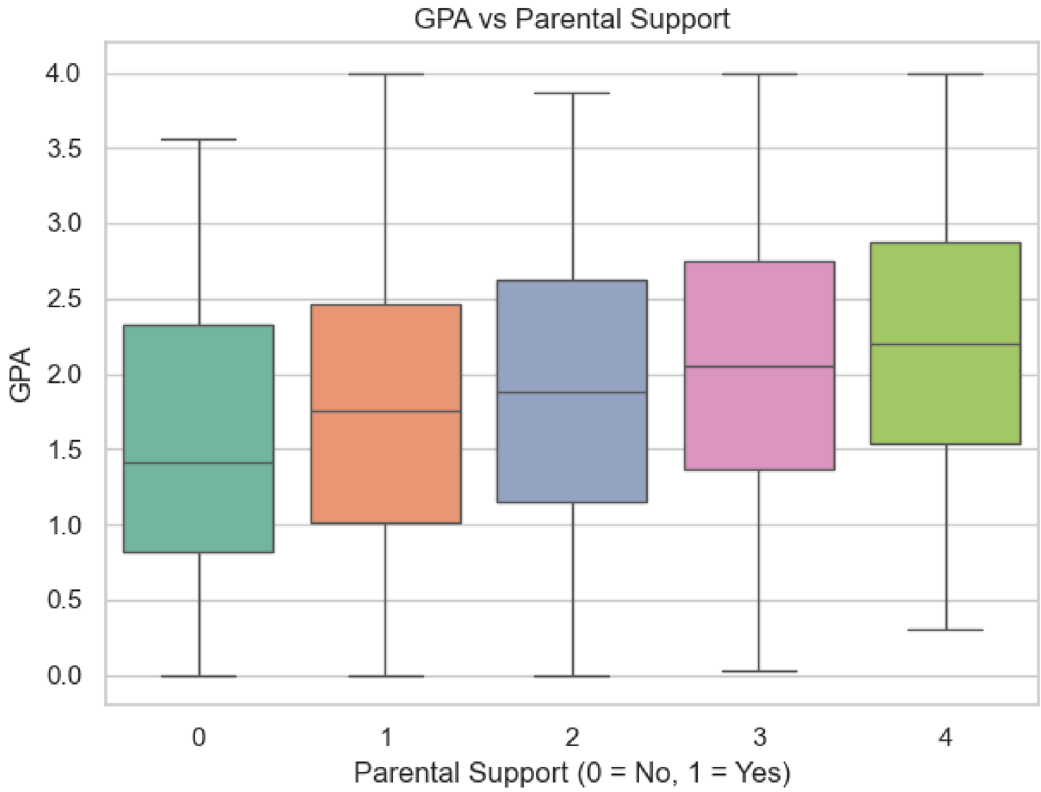
Correlational Heatmap (correlation of all features)



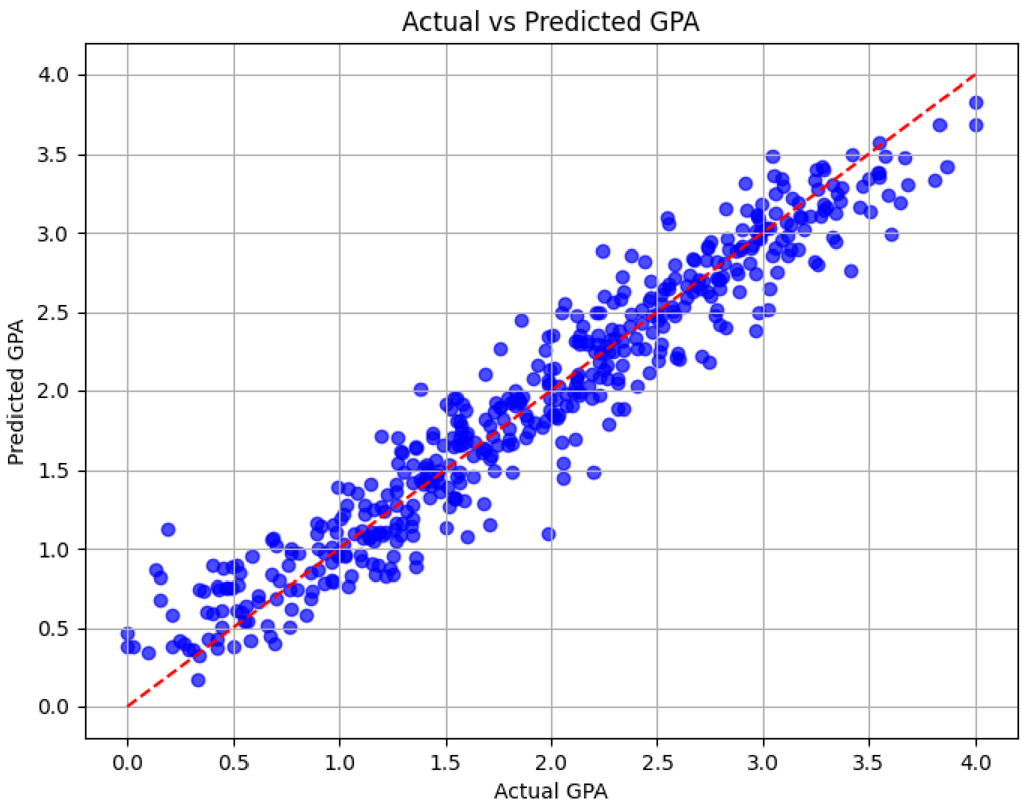
GPA by Gender



GPA by Parental Support



Actual GPA vs Predicted GPA (Scatter Plot)



Conclusion

The Student Performance Prediction project using Artificial Intelligence successfully demonstrates how data-driven models can provide actionable insights in the educational sector. By applying machine learning regression techniques, we were able to build a model that accurately predicts student GPA based on academic, demographic, and behavioural data.

The model's performance, as measured by various evaluation metrics, indicates that artificial intelligence can play a crucial role in identifying students who may be at risk of underperforming. This empowers educators, parents, and policymakers to implement timely and effective interventions that can significantly improve learning outcomes.

This project not only meets its objective of GPA prediction but also opens the door to a wide array of educational analytics applications. From real-time performance dashboards to smart counselling recommendations, the possibilities are vast.

Furthermore, the process of building this model—from data preprocessing and exploration to model training and prediction—serves as a practical blueprint for deploying AI in academic settings. It reinforces the idea that artificial intelligence, when used responsibly, can be a powerful tool in improving the quality and equity of education.

Future Work

While this project has laid a strong foundation, several improvements and expansions can enhance its accuracy and usability in real-world educational environments:

1. **Larger and More Diverse Datasets:** Incorporating datasets from different schools, regions, or even countries can help build a more generalized and robust model that works across diverse academic environments.
2. **Integration of Deep Learning Techniques:** Future versions of this model could leverage deep neural networks and ensemble methods to capture complex patterns in student behaviour and academic performance.
3. **Real-Time Performance Tracking:** Developing a web or mobile application where student data can be entered dynamically would make the system more practical and user-friendly. This could include dashboards for teachers, students, and parents.
4. **Feature Expansion:** Future work could include additional features such as classroom participation, assignment completion rates, and emotional well-being indicators, which could improve the predictive power of the model.
5. **Explainable AI (XAI):** Implementing techniques that make the model's predictions more interpretable would help in gaining trust among educators and policymakers.
6. **Policy-Level Integration:** With proven accuracy and fairness, such predictive systems could be integrated into academic policies for resource allocation, personalized learning plans, and performance monitoring.

By pursuing these directions, the project can evolve from a simple GPA prediction tool into a comprehensive AI-based student support system.